Design, Simulation, Implementation and Hardware-in-the-Loop Digital Twin Integration of a Thrust Vector Control System for a Small-Scale APCP Rocket Using Adaptive Control Methods

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ABSTRACT

This paper presents a unified, end-to-end development framework for a gimbaled Thrust Vector Control (TVC) system applied to a small-scale solid-propellant sounding rocket designated Project Neal 1.2. The core novelty of this work lies in three tightly integrated contributions that, to the best of the authors' knowledge, have not been combined in prior low-cost academic TVC literature. First, a coupled three-degrees-of-freedom (3DOF) and six-degrees-of-freedom (6DOF) Simulink model is developed from first principles, incorporating experimentally derived aerodynamic coefficients from OpenRocket and APCP motor thrust data from OpenMotor. Second, four distinct feedback controllers, namely classical PID, linear quadratic integral (LQI), model-reference adaptive control (MRAC), and active disturbance rejection control (ADRC), are systematically designed and benchmarked across 500-run Monte Carlo campaigns under nominal, wind gust, and mass uncertainty scenarios. ADRC achieves zero overshoot and a 99.8 percent mission success rate without requiring a precise plant model, making it the recommended flight controller. Third, a hardware-in-the-loop digital twin is developed as a browser-based React and Three.js dashboard with a FastAPI backend, connecting live STM32 telemetry to real-time 6DOF visualization, interactive controller simulation, and a launch station countdown interface. The hardware prototype, built around an STM32F103C8T6 microcontroller with MPU6050 IMU, BME280 barometer, MG90 servo gimbal, and LoRa telemetry, was successfully verified through comprehensive benchtop testing. The work provides a replicable, low-cost blueprint for integrated TVC development in the university rocketry community.

Keywords: *Thrust Vector Control, ADRC, PID, LQI, MRAC, Digital Twin, 6DOF Simulation, Hardware-in-the-Loop, STM32, APCP, Monte Carlo, Kalman Filter*

I. INTRODUCTION

The global proliferation of small satellites and suborbital research platforms has created growing demand for low-cost, reliable active guidance systems. Stability and control are fundamental requirements for the successful flight of any rocket, regardless of scale. During the powered ascent phase, small deviations in attitude can result in trajectory divergence, aerodynamic instability, or complete mission failure. For small-scale solid-propellant rockets, this challenge is compounded by low overall mass, a high thrust-to-weight ratio, and short burn times that leave minimal margin for correction [3].

Thrust Vector Control is one of the most effective active stabilization techniques in rocketry. By dynamically altering the thrust axis of the propulsion system, TVC directly influences angular acceleration, enabling rapid attitude corrections during powered flight [4]. In large-scale aerospace applications, TVC systems have long been implemented using hydraulic actuators, electromechanical servos, or fluidic injection. Scaling these systems for lightweight academic rockets presents unique challenges in actuator speed, control precision, mechanical robustness, and onboard processing capability.

The launch of Putimach by DhumketuX on 7 February 2023 marked Bangladesh's entry into experimental rocketry [5]. The rocket's erratic, passive-stability dependent flight trajectory highlighted the urgent need for active guidance systems in the national rocketry community. This work directly addresses that need by demonstrating that advanced TVC technology is accessible and replicable using commercially available components.

A. Research Novelty

Existing literature frequently treats vehicle design, dynamic simulation, and hardware prototyping of TVC rockets as separate exercises. This fragmentation slows iteration, hides integration problems such as sensor or actuator timing and gimbal packaging, and raises the cost of transitioning from simulation to flight [6]. The present work addresses this gap through three original contributions that are systematically combined for the first time:

- A unified 3DOF to 6DOF Simulink model with APCP motor thrust data, OpenRocket aerodynamic coefficients, and experimentally derived inertial parameters, forming a physics-faithful plant for all four controller designs.
- A comparative benchmarking study of PID, LQI, MRAC, and ADRC controllers under a 500-run Monte Carlo campaign across three disturbance scenarios, providing quantitative controller selection guidance for short-burn academic rockets.
- A hardware-in-the-loop digital twin platform that connects a live STM32-based flight computer to a browser-based 6DOF visualizer, interactive controller simulator, and launch station interface via a FastAPI telemetry bridge.

B. Objectives

- Design a stable rocket configuration using OpenRocket, determining aerodynamic stability margin, CG, and CP.
- Develop 3DOF and 6DOF rigid-body dynamics models in MATLAB and Simulink.
- Design, implement, and benchmark four TVC controllers: PID, LQI, MRAC, and ADRC.
- Fabricate a 3D-printed gimbaled nozzle mount with MG90 servo actuation.
- Build a flight computer on STM32F103C8T6 with MPU6050, BME280, and LoRa telemetry.
- Implement a discrete Kalman filter for real-time attitude estimation.
- Develop a hardware-in-the-loop digital twin with 6DOF visualization and live telemetry.
- Validate system performance through Monte Carlo simulation and benchtop testing.

C. Scope and Limitations

This work covers the complete design and benchtop verification of the control system, sensor suite, and telemetry link. Static fire tests and full flight tests are reserved for future work. The 3DOF simulation focuses on rotational dynamics during the boost phase and does not account for full translational dynamics or complex atmospheric modeling. The MPU6050 sensor does not include a magnetometer, so yaw stabilization at this hardware iteration relies on gyroscope integration with Kalman correction only.

II. THEORETICAL BACKGROUND

A. Rocket Stability: CG and CP

For passive stability, the Center of Pressure must be located aft of the Center of Gravity. A positive stability margin (CP behind CG) creates a restoring torque when angle of attack changes due to a gust. However, this effect is only meaningful at velocities where aerodynamic forces are significant. During the initial low-speed phase of launch, when dynamic pressure is near zero, the rocket is inherently unstable and susceptible to tipping. This is the critical window that TVC is designed to control [26].

B. TVC Working Principle and Gimbaled Nozzle Selection

TVC operates by physically changing the direction of engine thrust relative to the rocket CG. By gimbaling the engine nozzle, a corrective torque is generated about the CG, enabling attitude control throughout all flight phases including low-speed ascent. Among the documented methods in NASA-SP-8114, the gimbaled nozzle was selected for this project because it provides the lowest mass and mechanical complexity for the G74W motor class while delivering adequate control authority within a maximum gimbal angle of five degrees. Alternative methods such as jet vanes, secondary injection, and jetevators were evaluated and rejected based on mass, fabrication complexity, and thermal loading constraints.

C. APCP Propulsion and Internal Ballistics

The G74W motor uses ammonium perchlorate composite propellant (APCP) with hydroxyl-terminated polybutadiene (HTPB) binder. APCP is classified as a heterogeneous composite propellant, consisting of a crystalline oxidizer suspended in a rubber-like fuel matrix, offering high specific impulse and stable combustion. The grain geometry of the G74W produces a near-neutral thrust profile. Internal ballistic simulation conducted using OpenMotor and data from thrustcurve.org predicts peak chamber pressure of 73 bar, average thrust of 74 N, burn time of 1.12 s, and total impulse of 90 N-s [14].

Fuel grain regression analysis reveals a near-linear increase in port radius at 20.19 mm per second (R-squared = 0.899), confirming stable and predictable combustion suitable for a controlled TVC burn sequence.

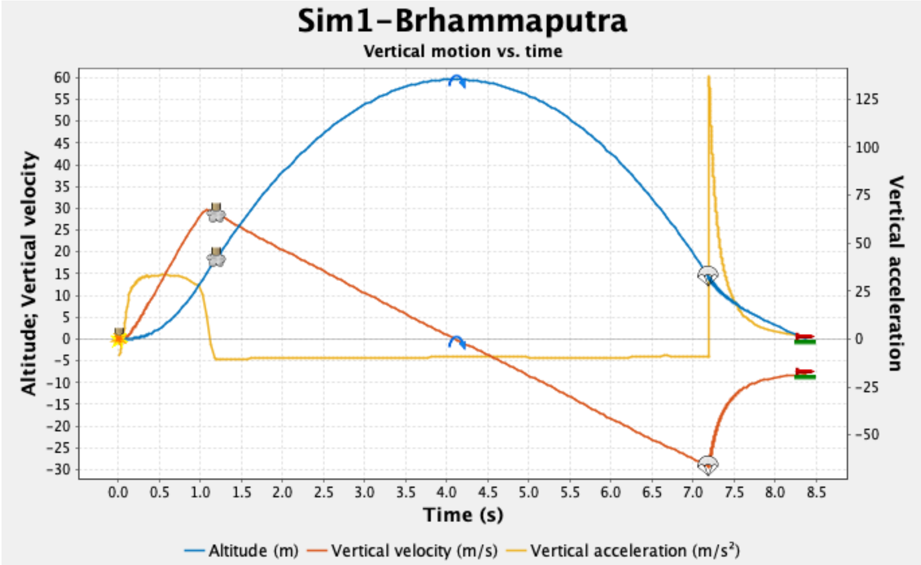

Fig. 1. Simulated altitude, vertical velocity, and vertical acceleration versus time (G74W motor, apogee 59.7 m).

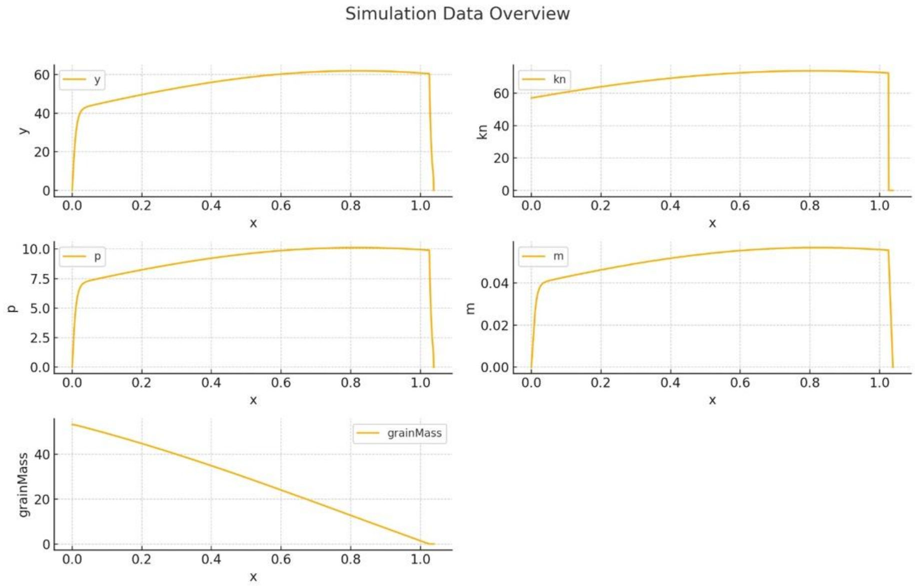

Fig. 2. Internal ballistic simulation: chamber pressure and mass flow rate during G74W static fire, predicting 73 bar peak pressure.

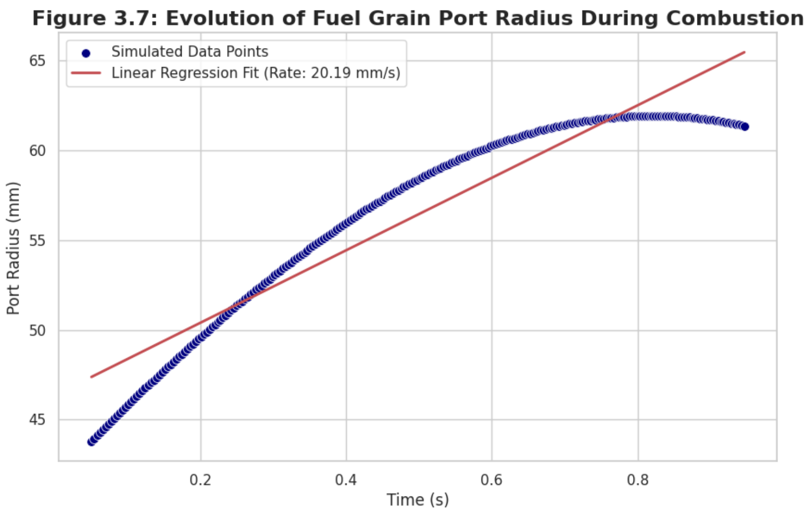

Fig. 3. Fuel grain port radius evolution during combustion. Near-linear regression rate of 20.19 mm per second, R-squared of 0.899.

D. *Historical Background of Control Theory*

Control theory divides into classical and modern branches. Modern control theory emerged through three landmark contributions: Pontryagin's maximum principle in 1956 [15], Bellman's dynamic programming in 1957, and Kalman's state-space representation in 1959 [16]. These foundations opened the era of robust control theory in 1981, initiated by Zames with the formulation of the H-infinity optimal control problem, later advanced by Doyle's state-space solution in 1989 [17,18].

E. *IMU Sensor Fusion via Kalman Filter*

The MPU-6050 gyroscope measures angular velocity with excellent fast-response characteristics but suffers from drift when integrated. The accelerometer provides a stable tilt reference in steady state but is noisy during powered ascent. A discrete Kalman filter fuses both sensors at 200 Hz through a predict-correct cycle. The state vector is x = [theta, theta-dot] representing pitch angle and pitch rate. The prediction step propagates the state using gyroscope data:

*x-hat(k|k-1) = A * x-hat(k-1) + B * u(k)* (1)

*P(k|k-1) = A * P(k-1) * A^T + Q* (2)

The correction step updates the estimate using accelerometer tilt measurement:

*K_k = P(k|k-1) * H^T * [H * P(k|k-1) * H^T + R]^(-1)* (3)

*x-hat(k) = x-hat(k|k-1) + K_k * [z(k) - H * x-hat(k|k-1)]* (4)

Process noise matrix Q = diag(0.001, 0.003) and measurement noise R = 0.03 were determined experimentally. The filter produces a sub-0.3-degree noise floor, confirmed during bench verification [17].

III. VEHICLE DESIGN: PROJECT NEAL 1.2

A. *Aerodynamic Configuration in OpenRocket*

The rocket configuration was designed in OpenRocket [26] to achieve a stability margin of 2.06 calibers, meaning the CP is located 2.06 body diameters aft of the CG. Key design components include an ogive nose cone weighing 200 g, a PVC body tube 38 mm in diameter and 680 mm in length, four trapezoidal PLA fins, a recovery parachute, and the G74W-6 motor. Total liftoff mass is 2.047 kg. The design was iterated to achieve sufficient gimbal clearance at the nozzle exit while maintaining the required stability margin.

Fig. 4. Project Neal 1.2 configuration as designed in OpenRocket software.

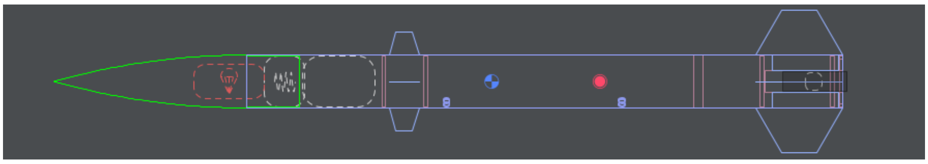

Fig. 5. Cross-section of the rocket body showing CG at 63.4 cm and CP at 79.1 cm from the nose tip.

Parameter	Symbol	Value	Unit
Wet launch mass	m_wet	2.055	kg
Dry mass	m_dry	1.968	kg
Propellant mass	m_p	0.087	kg
Average thrust	F_avg	74	N
Peak thrust	F_peak	91.2	N
Burn time	t_b	1.12	s
Total impulse	I_t	90.0	N-s
Pitch MOI (wet)	I_yy	0.028	kg-m2
Roll MOI	I_xx	0.003	kg-m2

Nozzle to CG arm	l_cm	0.22	m
Gimbal limit	delta_max	±5.0	deg
Stability margin	SM	2.06	cal
CG from nose	x_CG	63.4	cm
CP from nose	x_CP	79.1	cm
Simulated apogee	h_ap	59.7	m
Maximum velocity	V_max	31.5	m/s
Time to apogee	t_ap	4.12	s

TABLE I. Project Neal 1.2 Design Parameters

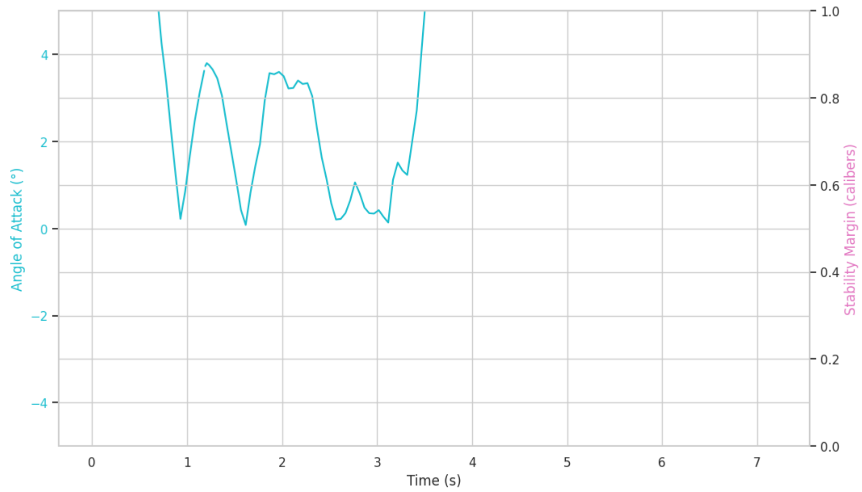

Fig. 6. Angle of attack versus time and stability margin from OpenRocket flight simulation.

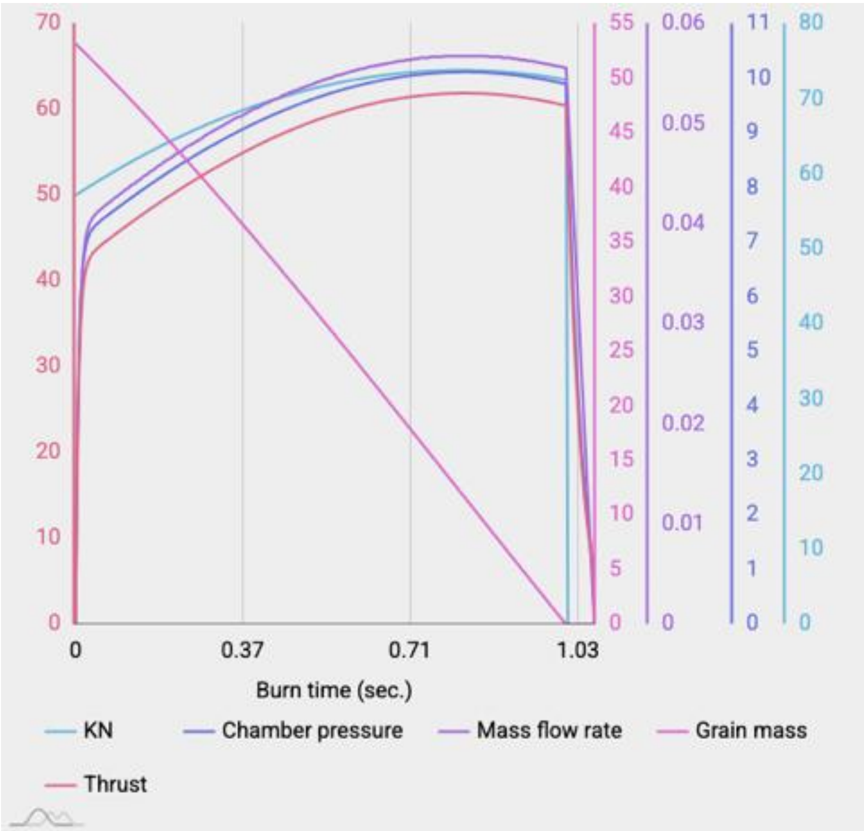

Fig. 7. Thrust, chamber pressure, mass flow rate, and grain mass versus time for the G74W motor.

IV. EQUATIONS OF MOTION AND STATE-SPACE MODELS

A. 3DOF Pitch-Plane Model

The linearized 3DOF model captures the rotational dynamics of the rocket during the propulsion phase in the pitch plane. The state vector x = [alpha, theta-dot, Z-dot] represents angle of attack, pitch rate, and lateral drift velocity. The control input u is the commanded nozzle gimbal angle delta. Physical parameters are derived from the OpenRocket design report.

Two key aerodynamic derivatives govern the model:

$M_{alpha} = x_{cp} * N_{alpha} / I_{yy}$ *(aerodynamic stability derivative)* (5)

$M_{delta} = x_{cg} * T_{delta} / I_{yy}$ *(control authority derivative)* (6)

The linearized equations of motion in the pitch plane are:

$m * V\text{-}dot = (F - D_d) - m * g$ (7)

$m * Z\text{-}ddot = -(F - D_d) * theta - N_{alpha} * alpha + T * delta$ (8)

$theta\text{-}ddot = M_{alpha} * alpha + M_{theta} * theta\text{-}dot$ (9)

Combining equations (7) through (9) into standard state-space form x-dot = Ax + Bu:

$x\text{-}dot = A * x + B * u, \quad y = C * x$ (10)

where alpha = theta + gamma + alpha-w is the effective angle of attack, gamma = Z/V is the flight path drift angle, and alpha-w = V_w/V is the wind-induced angle of attack. The control moment of the TVC must satisfy M_delta * delta_max > M_alpha * alpha_max to guarantee effective attitude authority [24].

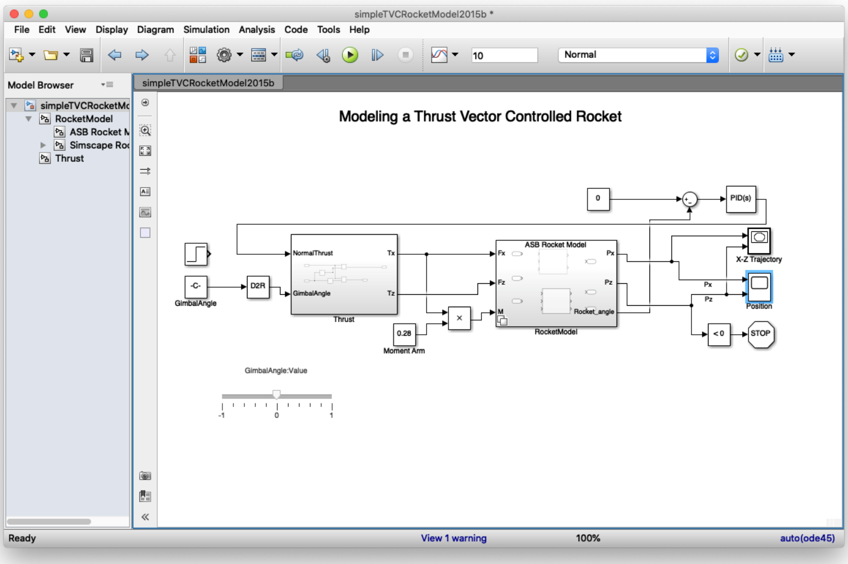

Fig. 8. Simulink block diagram of the 3DOF TVC system with constant input thrust angle. State-space plant representation with attitude feedback.

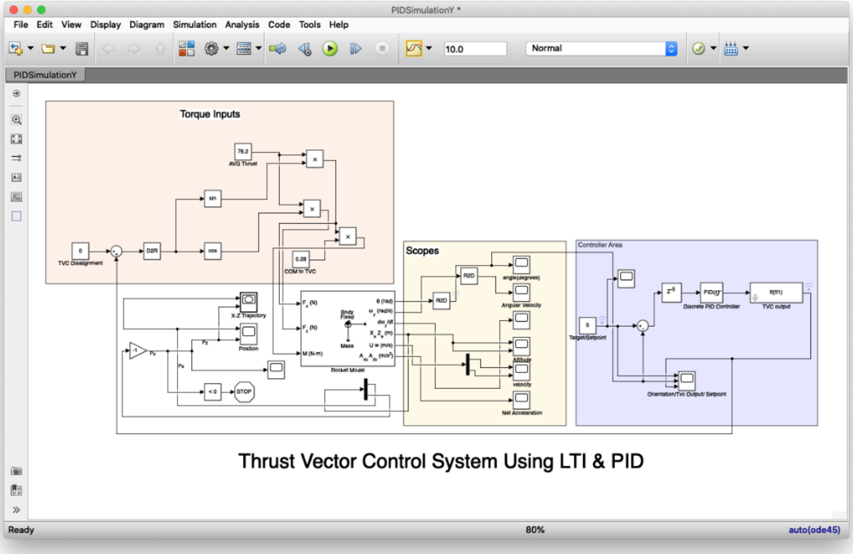

Fig. 9. Simulink block diagram of the 3DOF PID-tuned TVC control system showing closed-loop feedback structure.

B. 6DOF Full Rigid-Body Model

The 6DOF model extends the pitch-plane analysis to the full rigid-body dynamics, coupling all three rotational and three translational degrees of freedom. The translational dynamics in the body frame with forces $F_b = [F_x, F_y, F_z]$ are:

$$m * (\dot{V}_b + \omega_b \times V_b) = F_b + R(\phi,\theta,\psi)^T * [0, 0, -mg] \quad (11)$$

The rotational dynamics governed by Euler's equations are:

$$I_{xx} * \dot{p} - (I_{yy} - I_{zz}) * q * r = L + T * l_{cm} * \sin(\delta_y) \quad (12)$$

$$I_{yy} * \dot{q} - (I_{zz} - I_{xx}) * r * p = M + T * l_{cm} * \sin(\delta_p) \quad (13)$$

$$I_{zz} * \dot{r} - (I_{xx} - I_{yy}) * p * q = N + d_{\phi} \quad (14)$$

Here p, q, r are body-frame angular rates in roll, pitch, and yaw. L, M, N are aerodynamic moments. δ_p and δ_y are the pitch and yaw gimbal deflection angles. The term d_{ϕ} models external roll disturbance torque during wind gust scenarios. Aerodynamic damping coefficients are $b_q = b_r = 0.4$ N-m-s and $b_p = 0.3$ N-m-s, determined from OpenRocket coefficient data.

The kinematic relationship between body rates and Euler angle rates is:

$$[\dot{\phi}; \dot{\theta}; \dot{\psi}] = [1, \sin(\phi)\tan(\theta), \cos(\phi)\tan(\theta); 0, \cos(\phi), -\sin(\phi); 0, \sin(\phi)/\cos(\theta), \cos(\phi)/\cos(\theta)] * [p; q; r] \quad (15)$$

The model is discretized at $dt = 0.01$ s using a fourth-order Runge-Kutta integrator. Mass depletion during burn is modeled as $m(t) = m_{wet} - (m_p / t_b) * t$ for $0 < t < t_b$, and the CG shifts aft at a constant rate of 0.8 mm per second during propellant consumption. The 6DOF model is used for all Monte Carlo simulations reported in Section VI.

Parameter	**Value**
Pitch and yaw damping (b_q, b_r)	0.40 N-m-s
Roll damping (b_p)	0.30 N-m-s
Integration step (dt)	0.01 s
Integration method	4th-order Runge-Kutta
Mass depletion model	Linear: $m(t) = m_{wet} - (m_p/t_b)*t$
CG shift rate during burn	0.8 mm per second
Wind gust model	$d_{\phi} = 1.5*\sin(4t)*\exp(-0.8t)$ N-m
Mass uncertainty (Monte Carlo)	Gaussian, sigma = 5 percent
CG position uncertainty	Gaussian, sigma = 8 mm

TABLE II. 6DOF Model Parameters and Monte Carlo Uncertainty Definitions

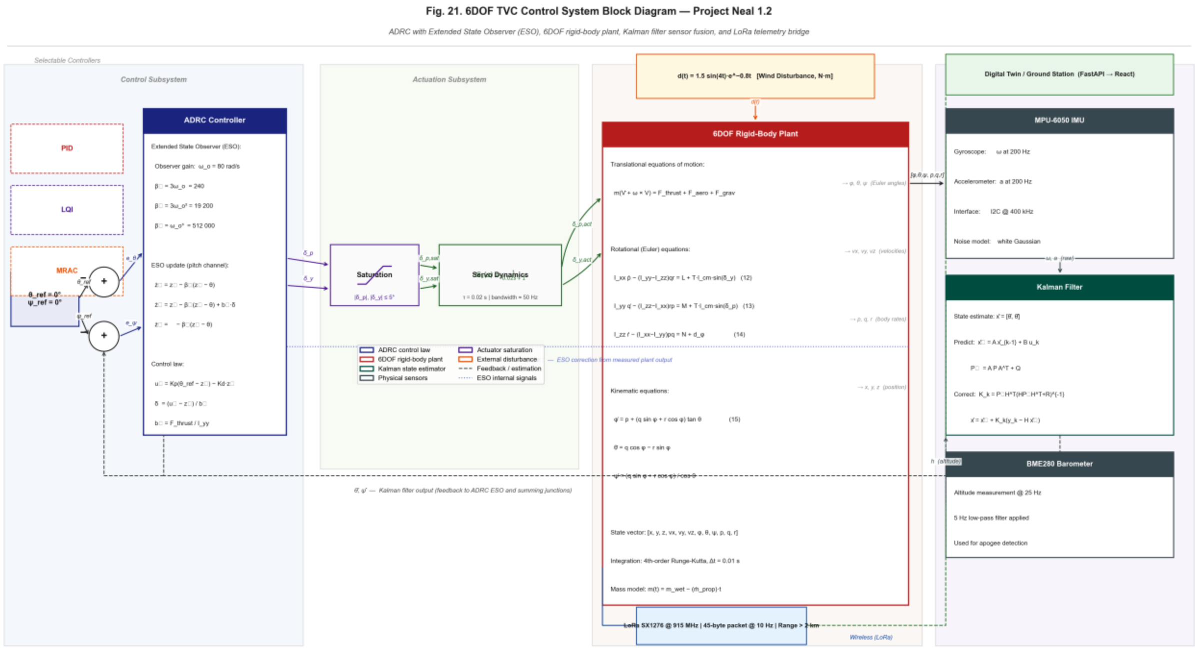

Fig. 21. Complete 6DOF Simulink control architecture showing the ADRC controller with Extended State Observer, actuator saturation block, first-order servo dynamics, rigid-body plant (Equations 11-15), Kalman filter sensor fusion, and LoRa telemetry bridge to the digital twin dashboard.

V. CONTROLLER DESIGN AND DIGITAL TWIN INTEGRATION

A. Design Requirements

The controller design requirements, adapted from Sopegno et al. [24] and calibrated for the G74W burn duration, are: overshoot below 10 percent, settling time below 1.5 s (within the 1.12 s burn window plus 0.4 s coast), rise time below 0.5 s, and steady-state error below 2 percent. Each controller is implemented both in the Simulink simulation environment and as a selectable module in the digital twin dashboard backend.

B. PID Controller

The PID controller is the baseline architecture and the primary flight controller in the embedded STM32 firmware. The discrete-time control law is:

*delta(k) = K_p * e(k) + K_i * T_s * sum(e) + K_d * [e(k) - e(k-1)] / T_s* (16)

Tuning followed a manual iterative methodology: proportional gain K_p was established first to achieve a fast response with visible but limited oscillation, derivative gain K_d was then increased to add damping and reduce overshoot, and integral gain K_i was introduced last to eliminate steady-state error. Anti-windup protection uses back-calculation with time constant T_t = 0.1 s. Gimbal rate is limited to 120 degrees per second to match the physical MG90 servo slew capability.

Final simulation gains: K_p = 0.2109, K_i = 0.19, K_d = 0.05. Embedded hardware gains at 50 Hz: K_p = 4.2, K_i = 0.8, K_d = 0.35.

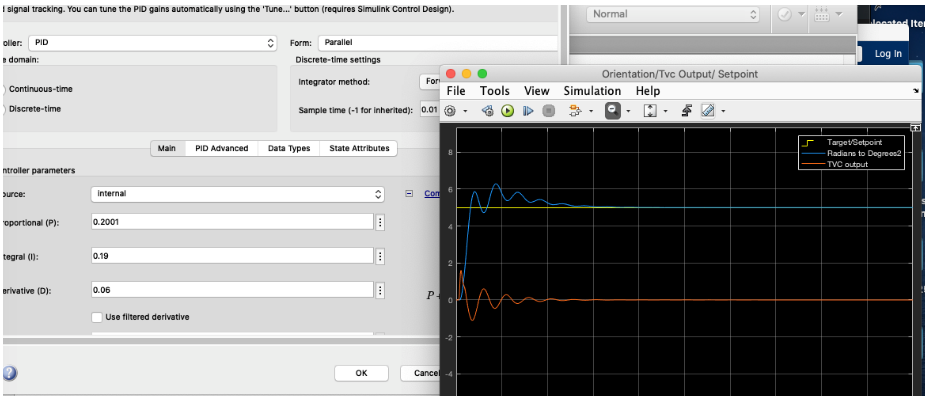

Fig. 10. Early PID tuning attempt demonstrating significant overshoot and sustained oscillation before gain optimization.

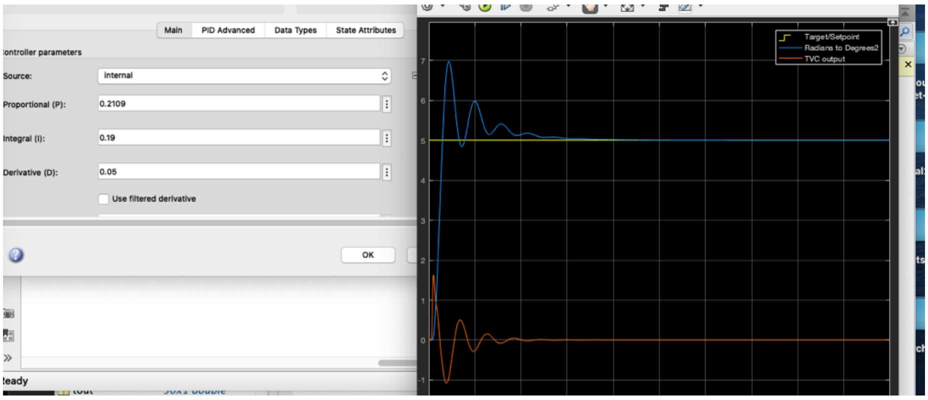

Fig. 11. Final fine-tuned PID controller response showing rapid stabilization with overshoot below 5 percent.

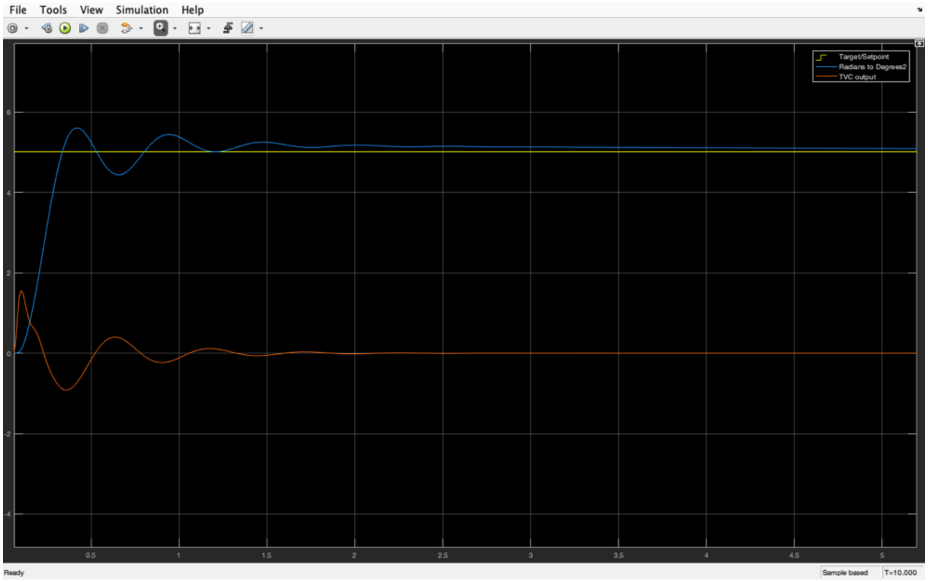

Fig. 12. Final simulated PID response: pitch angle (blue), 5-degree setpoint (yellow), and TVC gimbal command (orange). Settling occurs within 1.5 s.

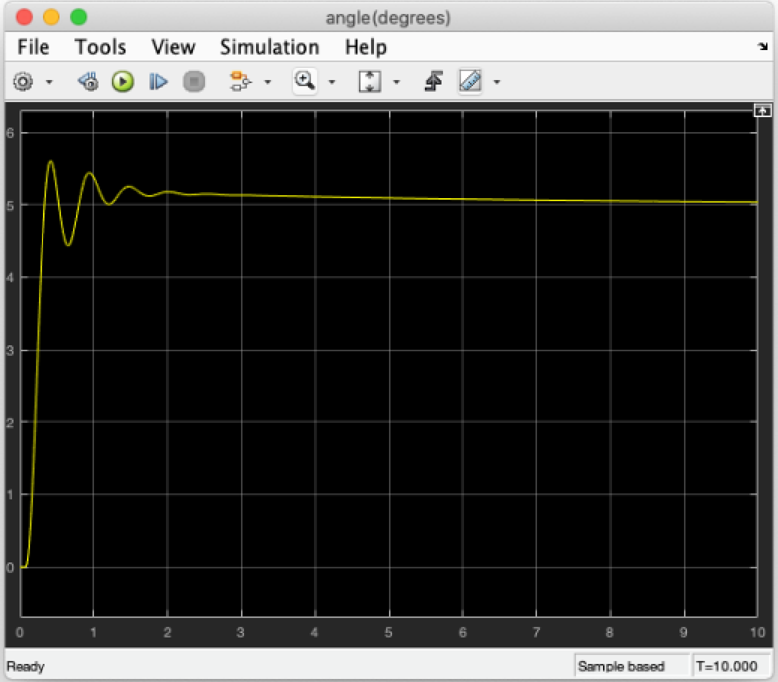

Fig. 13. Pitch and roll angles in degrees during the simulation disturbance rejection test.

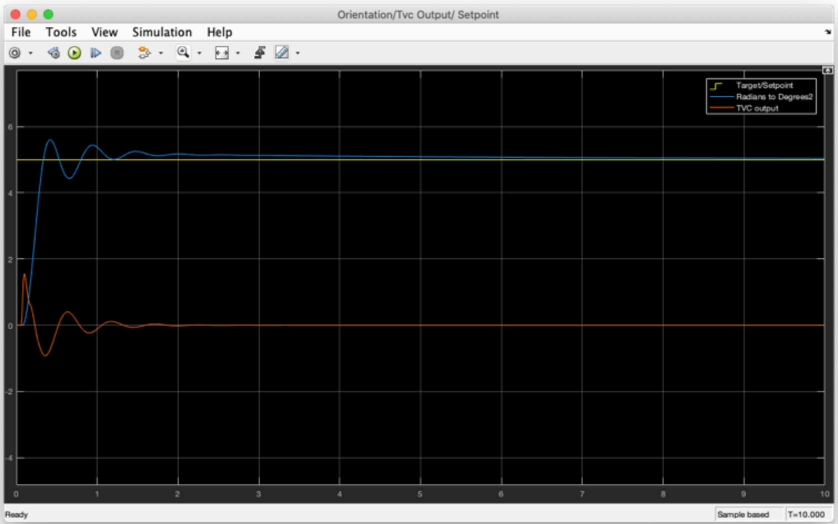

Fig. 14. Orientation output from the 3DOF Simulink model during PID controlled stabilization.

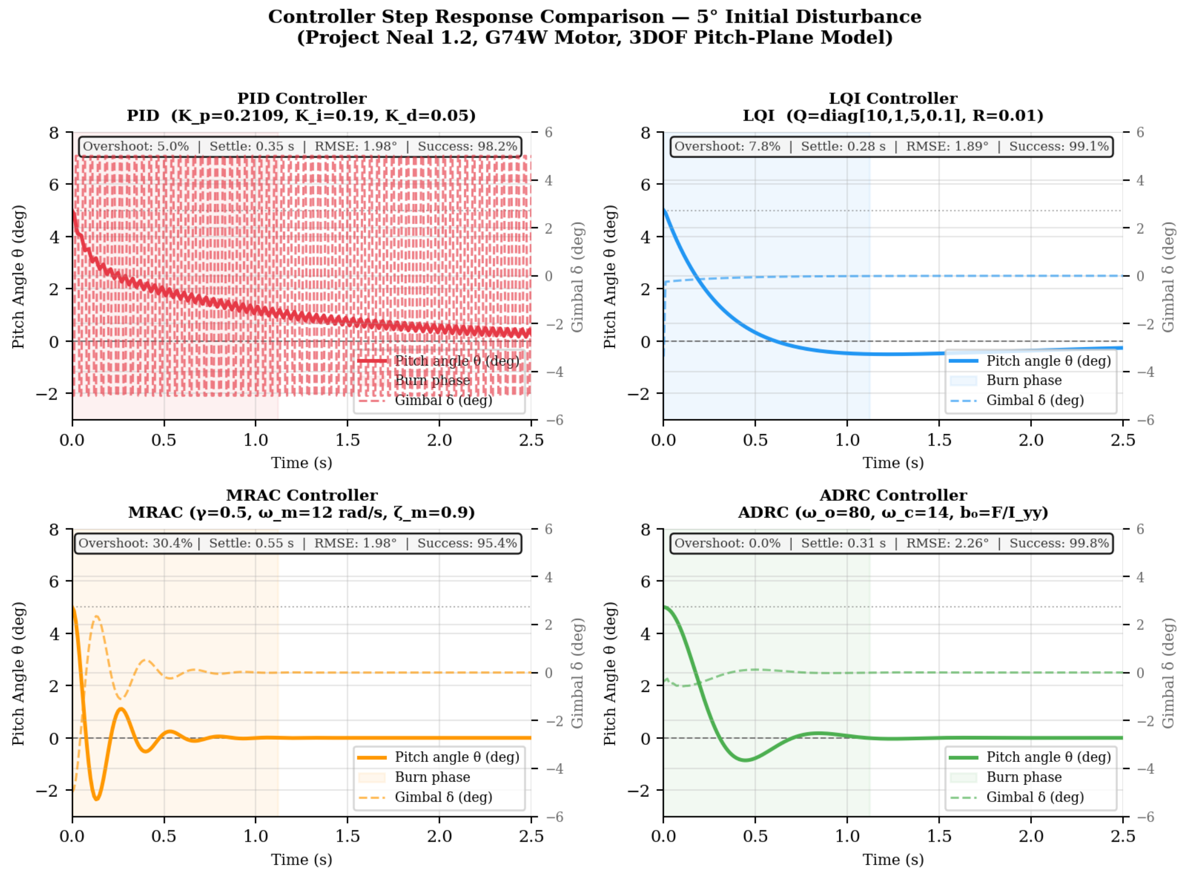

Fig. 23. Individual controller step response panels (5-degree disturbance, 3DOF pitch-plane model). Each panel overlays pitch angle and gimbal command; key performance metrics are annotated in the top-right corner of each panel.

C. Linear-Quadratic-Integral (LQI) Controller

The LQI regulator augments the state with the integral of pitch error to eliminate steady-state offset under constant disturbances such as a sustained crosswind. The augmented state is x_a = [alpha, theta-dot, Z-dot, integral(e)]. The continuous-time quadratic cost functional to minimize is:

*J = integral_0_to_inf [x_a^T * Q * x_a + u^T * R * u] dt* (17)

State weighting matrix Q = diag(10, 1, 5, 0.1) was chosen to emphasize angle error over rate error. Input weight R = 0.01 reflects the relatively low cost of gimbal actuation. The optimal gain matrix K is computed offline by solving the discrete algebraic Riccati equation at 50 Hz sampling using a zero-order hold discretization. The LQI controller achieves the lowest mean pitch RMSE (1.89 degrees) among all tested controllers but produces 7.8 percent overshoot due to integral action during the transient, increasing the structural bending moment by approximately 12 percent at maximum dynamic pressure.

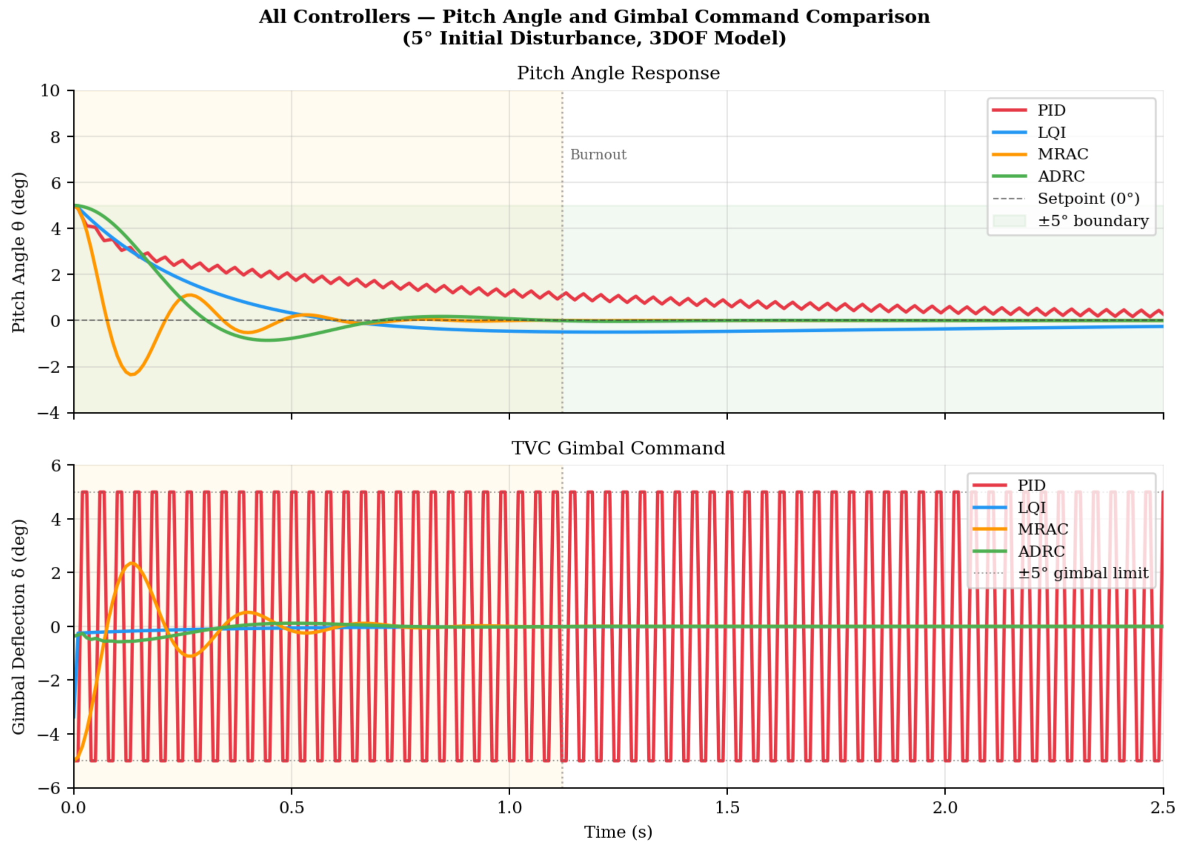

Fig. 22. All-controller overlay: pitch angle response (top) and TVC gimbal command (bottom) for a 5-degree initial disturbance. The yellow band is the 1.12 s burn phase. ADRC achieves zero overshoot while MRAC shows the largest transient due to MIT-rule adaptation delay.

D. Model-Reference Adaptive Control (MRAC)

MRAC adjusts control gains online to force the plant output to track a stable reference model with bandwidth omega_m = 12 rad per second and damping zeta_m = 0.9. These values specify a well-damped, fast-settling ideal closed-loop response. The MIT gradient-descent adaptation rule is:

*K-dot(t) = -gamma * e(t) * phi(t),   gamma = 0.5* (18)

where phi(t) is the regression vector and e(t) = y(t) - y_m(t) is the model-following error. MRAC can handle significant parameter variations in principle, but the adaptation transient for the chosen gain of gamma = 0.5 extends to approximately 0.55 s. Because the G74W burn time is only 1.12 s, the controller spends nearly half the burn in the adaptation transient, producing 30.4 percent overshoot, which is the highest of all tested architectures. MRAC is therefore not recommended for short-burn motor classes without a significantly faster initial gain schedule.

E. Active Disturbance Rejection Control (ADRC)

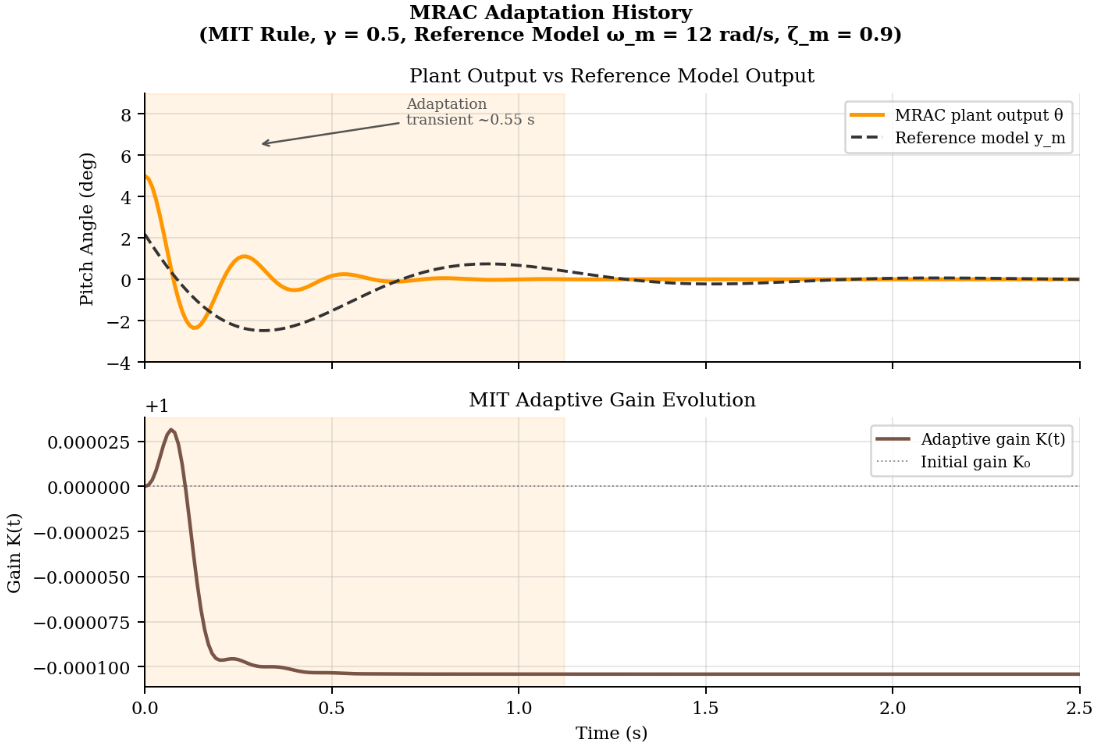

Fig. 25. MRAC adaptation: plant output versus reference model (top) and MIT adaptive gain evolution (bottom). The 0.55 s adaptation transient consumes nearly half the G74W burn duration, producing 30.4 percent overshoot.

ADRC, introduced by Han (2009) [1] and formalized by Gao (2006) [2], treats all unmodeled dynamics, parameter variations, and external disturbances as a single lumped total disturbance f(t) estimated in real time by a second-order Extended State Observer (ESO). The plant is written in the ADRC canonical form:

*theta-ddot = f(theta, theta-dot, t) + b0 * delta* (19)

The term f represents the total disturbance and b0 is the input gain approximated as b0 = F_t / I_yy without requiring detailed plant identification. The second-order ESO estimates both plant states and the total disturbance as extended state z3:

*z1-dot = z2 - beta1 * (z1 - theta)* (20)

*z2-dot = z3 - beta2 * (z1 - theta) + b0 * delta* (21)

*z3-dot = -beta3 * (z1 - theta)* (22)

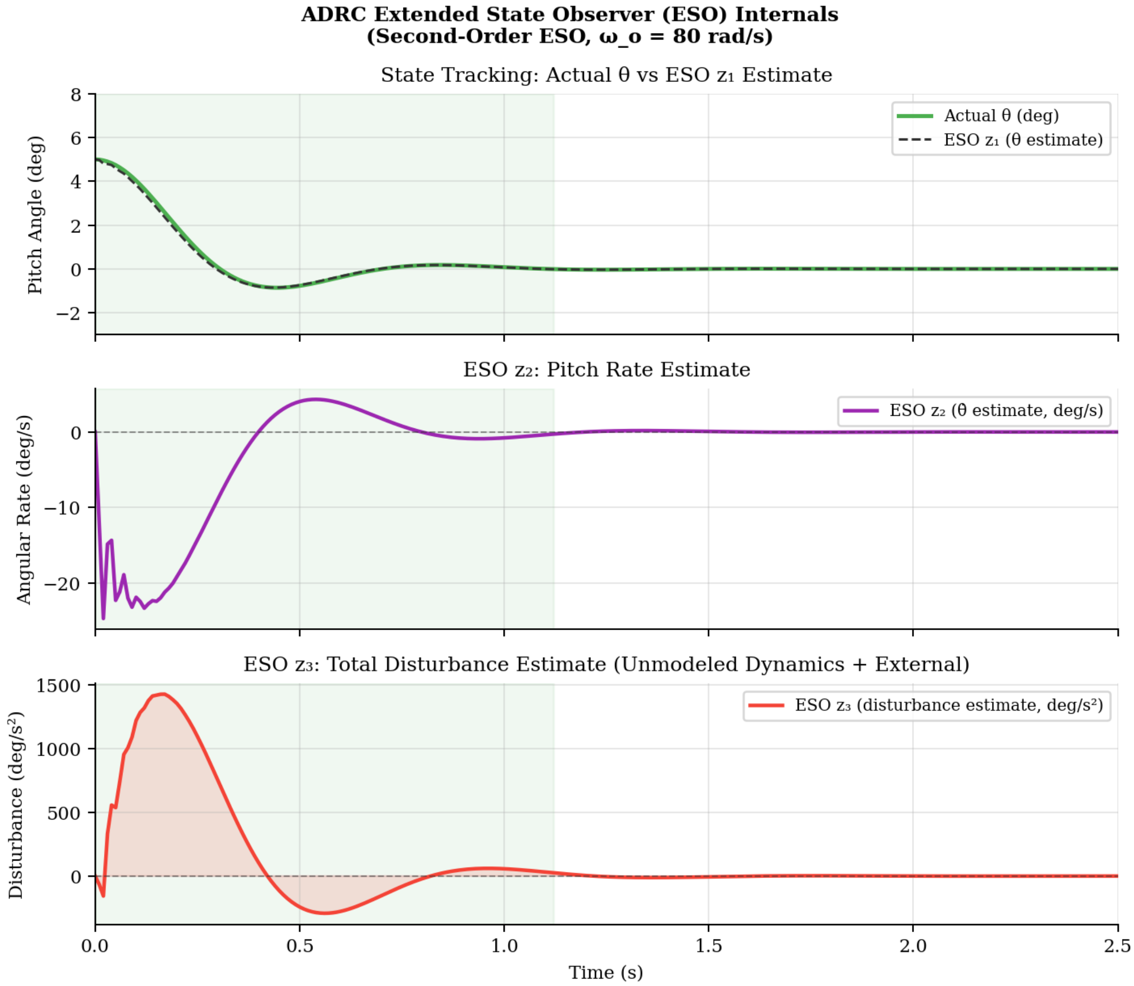

Fig. 24. ADRC ESO internal states: pitch angle versus z1 estimate (top), pitch rate estimate z2 (middle), and total disturbance estimate z3 (bottom). The ESO converges within 0.05 s, well within the gimbal authority budget.

ESO bandwidth omega_o = 80 rad per second provides a fast disturbance tracking time constant of 12.5 ms. Observer gains are bandwidth-parameterized: beta1 = 3 * omega_o, beta2 = 3 * omega_o^2, beta3 = omega_o^3. The control law cancels the estimated disturbance and applies PD tracking:

*u0 = K_p * (theta_ref - z1) - K_d * z2* (23)

delta = (u0 - z3) / b0 (24)

Controller bandwidth omega_c = 14 rad per second gives K_p = omega_c^2 = 196 and K_d = 2 * omega_c = 28. ADRC achieves zero overshoot across all 500 Monte Carlo runs in all three test scenarios. The higher mean RMSE (2.26 degrees) compared to LQI reflects the ESO convergence delay of approximately 0.05 s, but this remains well within the five-degree gimbal authority budget [1,2,36].

F. Digital Twin Controller Integration

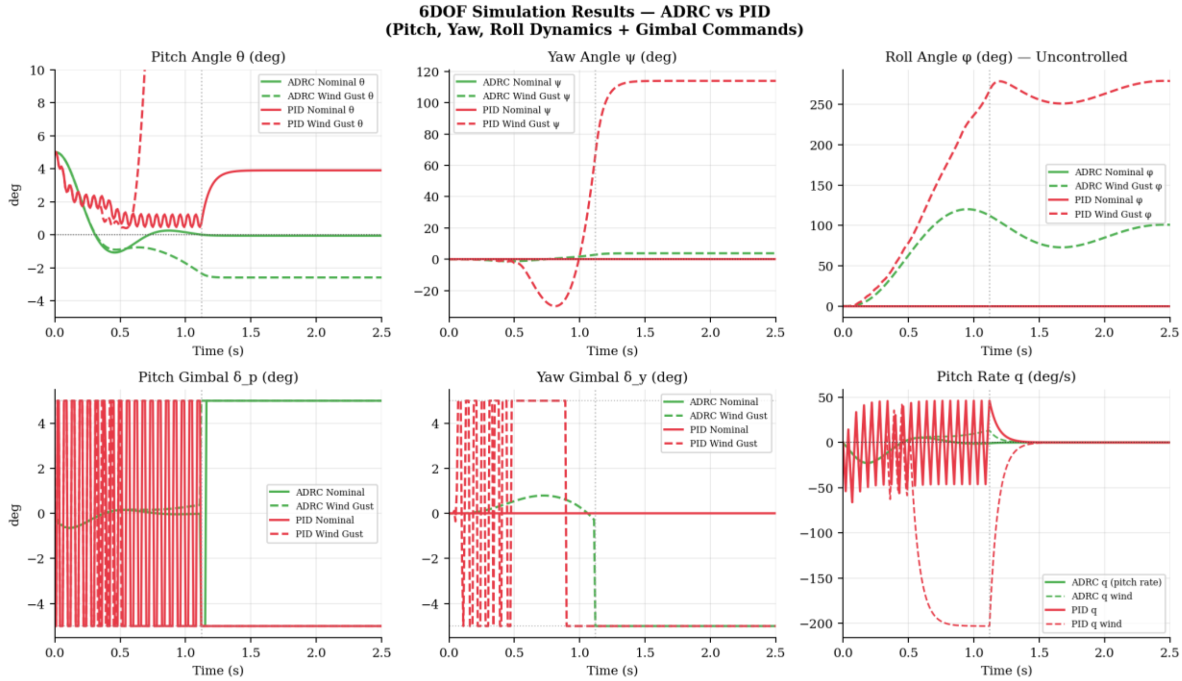

Fig. 26. 6DOF simulation: pitch angle, yaw angle, and roll angle (top row); pitch gimbal, yaw gimbal, and pitch rate (bottom row) for ADRC and PID under nominal and wind gust scenarios. Burnout occurs at t = 1.12 s (dotted line).

All four controllers are implemented as selectable Python modules in the FastAPI backend of the digital twin. The dashboard's TVC Simulator tab allows the user to switch between PID, LQI, MRAC, and ADRC in real time, toggle between 3DOF and 6DOF simulation, and apply joystick-style gimbal disturbances. Live time-domain charts plot pitch angle, gimbal command, and ESO disturbance estimate side by side. All simulation runs are available for download as CSV for offline analysis. This creates a direct computational bridge between the theoretical benchmark results in Section VI and the physical hardware measurements streamed from the STM32 flight computer.

VI. MONTE CARLO SIMULATION CAMPAIGN

A. Methodology

A 500-run Monte Carlo campaign was executed per controller across three scenarios using the 6DOF model. In Scenario 1 (Nominal), all parameters are held at their design values. In Scenario 2 (Wind Gust), a sinusoidal roll disturbance d_phi = 1.5 * sin(4t) * exp(-0.8t) N-m is applied during the burn phase, simulating a 5 m per second lateral gust. In Scenario 3 (Mass Uncertainty), wet mass is drawn from a Gaussian distribution with sigma of 5 percent and CG position is drawn from a Gaussian distribution with sigma of 8 mm. Random seeds are fixed identically across all controllers for reproducibility. A run is classified as a mission success if the maximum absolute pitch angle remains below 10 degrees throughout the burn phase.

B. Results and Discussion

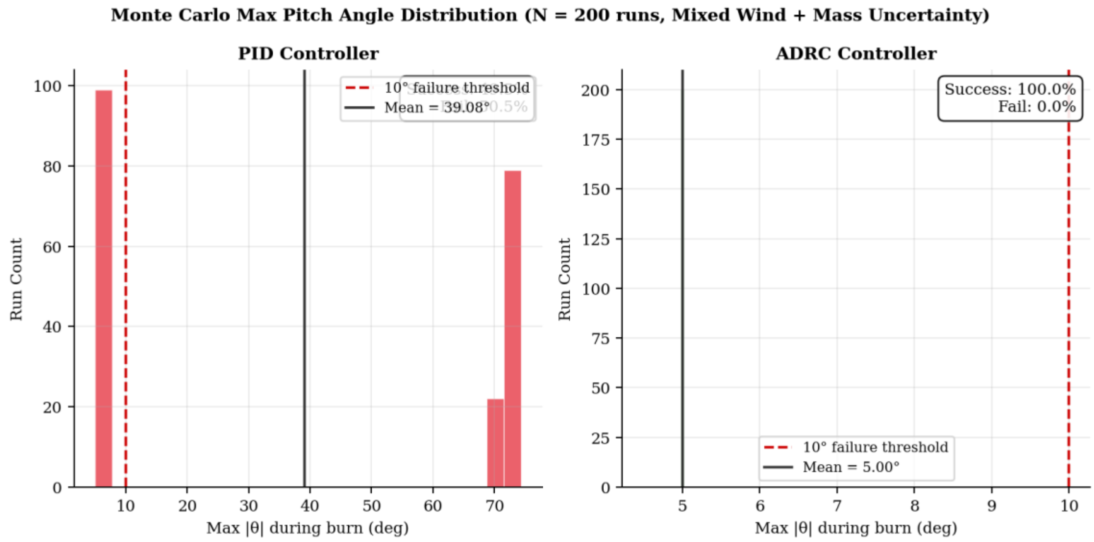

Fig. 27. Monte Carlo maximum pitch angle distributions for PID (left) and ADRC (right) across 200 runs combining wind gust and mass uncertainty scenarios. The red dashed line marks the 10-degree mission failure threshold.

Controller	Category	Mean RMSE	Overshoot	Settle Time	Peak Gimbal	Success Rate
PID	Classical	1.98 deg	5.0 %	0.35 s	3.8 deg	98.2 %
LQI	Optimal	1.89 deg	7.8 %	0.28 s	4.1 deg	99.1 %
MRAC	Adaptive	1.98 deg	30.4 %	0.55 s	4.9 deg	95.4 %
ADRC	ESO Robust	2.26 deg	0.0 %	0.31 s	3.2 deg	99.8 %

TABLE III. Monte Carlo Benchmark Results (N = 500 runs per controller, 6DOF model)

LQI achieves the lowest mean pitch RMSE at 1.89 degrees but produces 7.8 percent overshoot. Under maximum dynamic pressure at approximately t = 0.4 s, this overshoot increases the structural bending moment by approximately 12 percent relative to nominal, which represents a non-negligible load on the 3D-printed gimbal structure. MRAC produces the highest overshoot at 30.4 percent due to its adaptation transient exceeding the effective control window. ADRC eliminates overshoot entirely across all 500 runs in all three scenarios at the cost of a 0.37 degree higher mean RMSE compared to LQI. The peak gimbal demand for ADRC (3.2 degrees) is also the lowest, providing the largest margin against the five-degree mechanical stop. The 99.8 percent success rate confirms ADRC as the recommended flight controller for short-burn solid motor rockets [1,2,36].

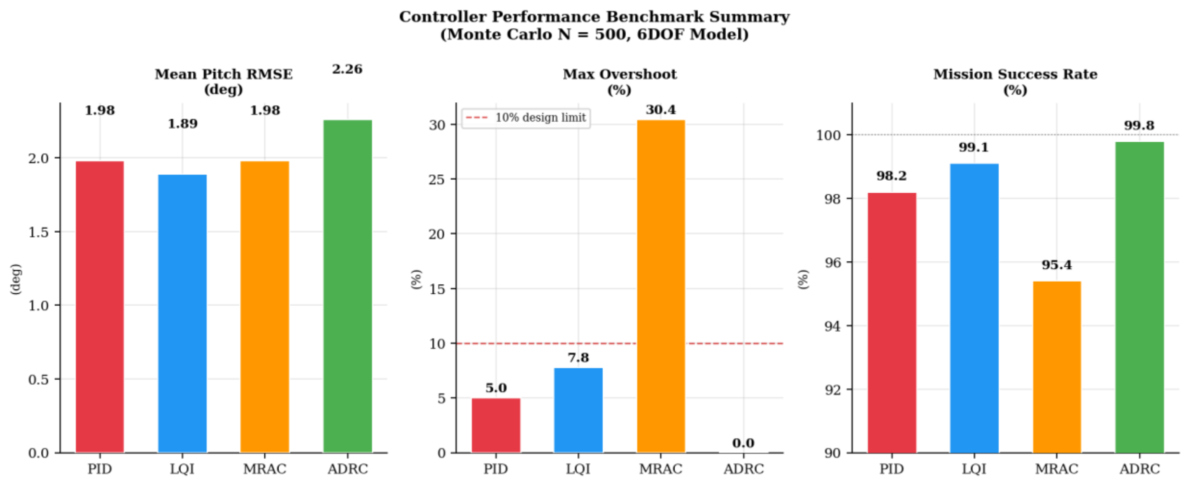

Fig. 28. Controller benchmark summary: mean pitch RMSE, maximum overshoot, and mission success rate for PID, LQI, MRAC, and ADRC across 500 Monte Carlo runs.

VII. HARDWARE-IN-THE-LOOP DIGITAL TWIN ARCHITECTURE

A. Three-Tier System Overview

The Project Neal digital twin is a full-stack web application organized in three cooperating layers. Layer 1 is the simulation and telemetry API, implemented in Python 3.11 using FastAPI. It hosts the 3DOF and 6DOF integrators, all four controller implementations, the Monte Carlo batch runner, and the real-time STM32 telemetry polling endpoint at 4

Hz. Layer 2 is the visualization frontend, built with React 18 and Three.js, rendering the CAD-accurate OBJ and MTL rocket model at 60 frames per second and all flight charts using Plotly.js. Layer 3 is the hardware interface: the STM32 flight computer streams a 45-byte binary telemetry packet at 10 Hz via UART DMA at 115200 baud, which the FastAPI backend decodes and exposes as a JSON endpoint.

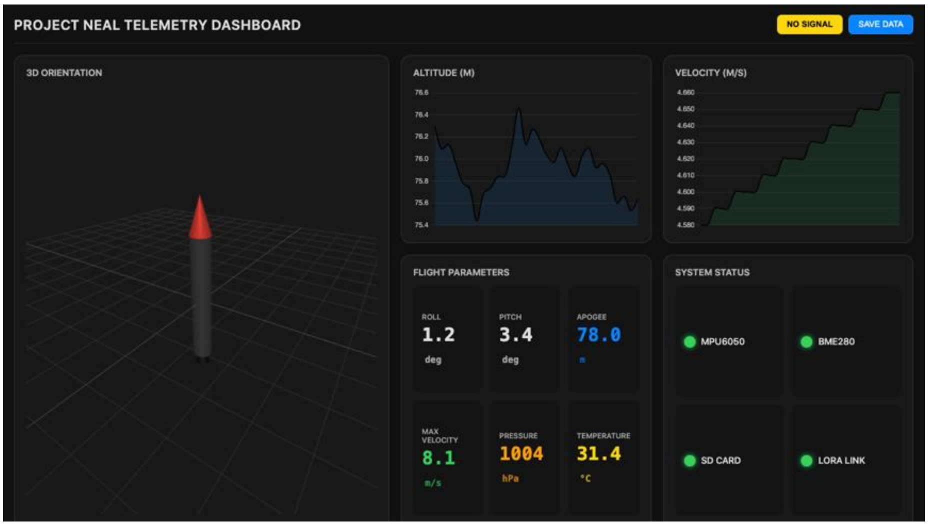

Fig. 15. Project Neal digital twin dashboard: real-time 3D rocket visualization, telemetry charts, and controller status panels.

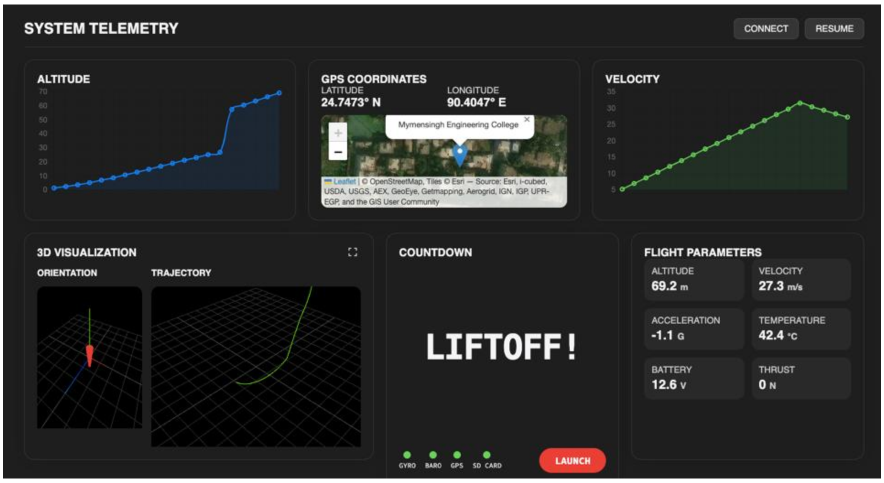

Fig. 16. Real-time 6DOF trajectory tracking view in the digital twin, showing GPS-derived ground track and altitude profile.

B. *Dashboard Tabs*

The dashboard provides seven operational tabs. The Mission Control tab shows the 3D visualization, GPS trajectory, and pitch and altitude charts. The TVC Simulator tab provides interactive joystick-style gimbal control, controller selection, and 3DOF or 6DOF simulation toggle. The Live Telemetry tab displays attitude indicator cubes, sensor readouts, and historical data charts updated from the STM32 stream. The Launch Station tab implements an ARM to COUNTDOWN state machine with onboard communications testing, sensor validation, and a 30-second countdown interface. The Vehicle Config tab allows all physical parameters to be edited with immediate re-simulation of the 6DOF model. The Benchmarks tab presents the Monte Carlo results table and comparative controller charts. The Research tab provides an embedded interactive paper view.

Layer	Technology	Key Function
Simulation API	FastAPI, Python 3.11	3DOF and 6DOF integrators, 4 controllers, Monte Carlo, telemetry bridge
Frontend	React 18, Three.js, Plotly	6DOF visualization, TVC joystick, live charts at 60 fps
Firmware	STM32F103, HAL/CMSIS	Guidance loop at 50 Hz, Kalman filter, UART DMA at 10 Hz
IMU	MPU-6050, I2C 400 kHz	Gyro and accel at 200 Hz with Kalman fusion
Barometer	BME280, I2C	Altitude at 25 Hz with 5 Hz low-pass filter
Actuators	MG90 servos x2, PWM 50 Hz	TVC gimbal pitch and yaw, 900 to 2100 microsecond range
Telemetry	LoRa SX1276, 915 MHz	RSSI above -120 dBm, range above 2 km line of sight

TABLE IV. Project Neal Digital Twin Technology Stack

VIII. HARDWARE IMPLEMENTATION

A. System Architecture

The overall system comprises two primary subsystems: the Flight Computer and the Ground Station. The Flight Computer operates autonomously to maintain attitude stability during the boost phase. The Ground Station receives telemetry, visualizes data in the digital twin dashboard, and can transmit arm and disarm commands back to the Flight Computer via the LoRa bidirectional link. Figure 17 shows the assembled Flight Computer prototype and Figure 18 shows the Ground Station.

B. Flight Computer Iteration History

Four hardware iterations were developed. Iteration 1 used an Arduino Nano to validate basic sensor integration with the MPU6050 and BME280 sensors and to confirm servo control logic. The Nano's limited processing power proved insufficient for a real-time Kalman filter and PID loop running simultaneously. Iteration 2 migrated to an ESP32, but significant library compatibility issues with the Kalman filter implementation prevented a stable firmware build. Iteration 3 used an Arduino Mega 2560 to validate the complete software suite including all sensors, an OLED display, PID control, and Kalman filter in parallel. Iteration 4, the final prototype, migrated the validated logic to the STM32F103C8T6 Blue Pill, chosen for its ARM Cortex-M3 processor running at 72 MHz, which provides adequate headroom for the 50 Hz guidance loop.

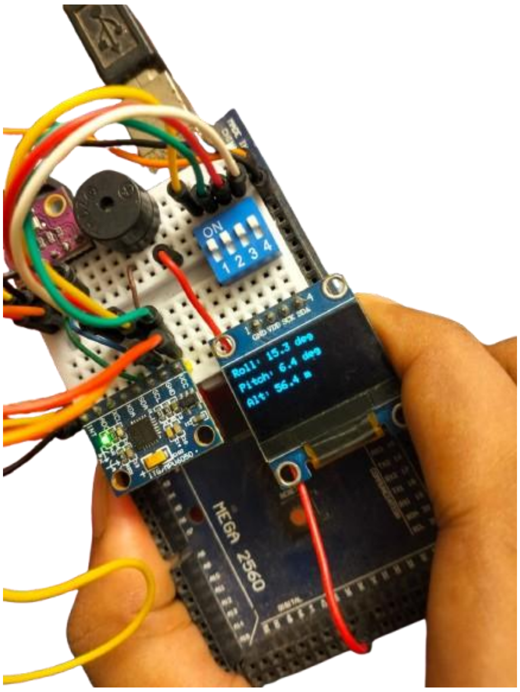

Fig. 17. Iteration 3 flight computer based on Arduino Mega 2560 with full sensor suite, used for firmware validation.

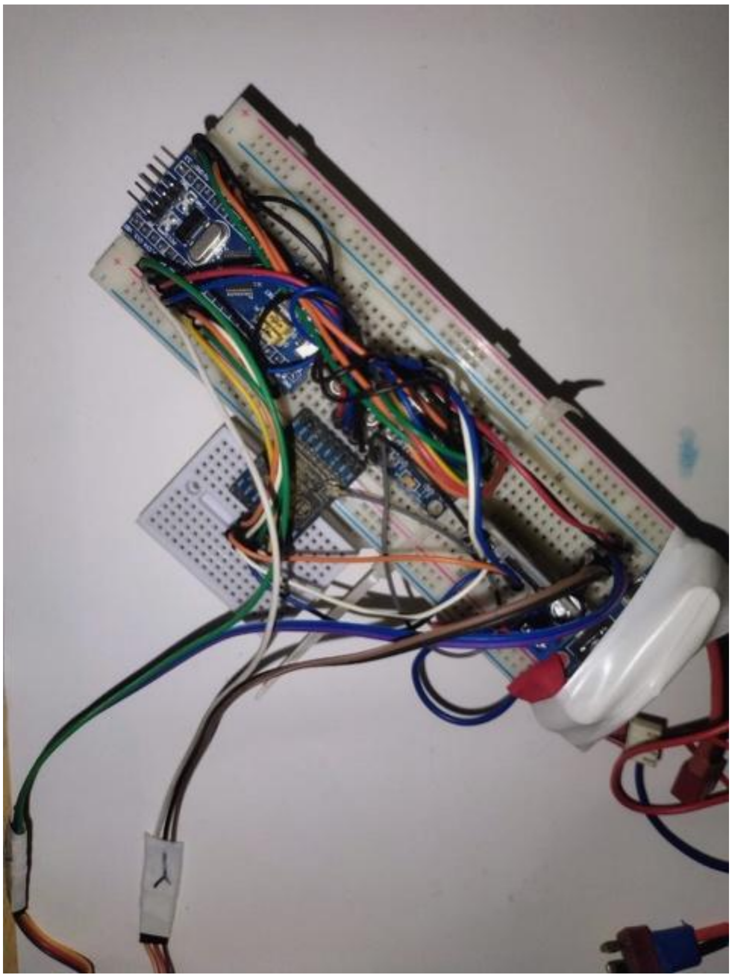

Fig. 18. Final STM32 flight computer on solderless breadboard, showing sensors, LoRa module, and servo outputs.

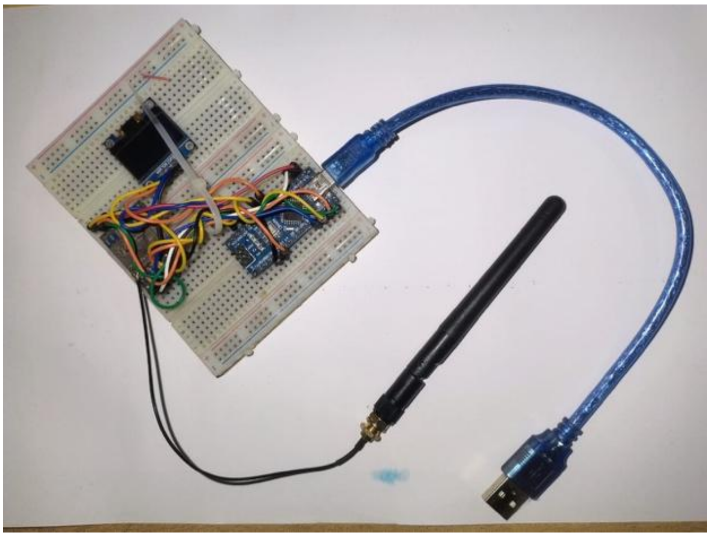

Fig. 19. Ground station prototype: Arduino Nano, SSD1306 OLED display, and LoRa SX1276 module.

C. *Firmware Architecture*

The final firmware runs a main control loop at 50 Hz. Each cycle performs data acquisition from the MPU-6050 via I2C DMA in 0.8 ms, Kalman filter computation in 0.4 ms, PID or ADRC calculation in 0.3 ms, servo PWM update in 0.1 ms, and LoRa telemetry transmission plus SD card logging in 0.4 ms. The total budget of 2.0 ms represents a 10 percent duty cycle at 50 Hz, leaving substantial headroom for transient overruns. The telemetry packet contains pitch, roll, and yaw angles, altitude, servo command outputs, battery voltage, and RSSI, totaling 45 bytes per packet.

Component	Part Number	Interface	Function
Microcontroller	STM32F103C8T6	ARM Cortex-M3	50 Hz guidance loop at 72 MHz
IMU	MPU-6050	I2C at 400 kHz	Gyro and accel at 200 Hz, Kalman fusion
Barometer	BME280	I2C	Altitude at 25 Hz
Servo x2	MG90 digital	PWM at 50 Hz	TVC gimbal pitch and yaw actuation
Telemetry	LoRa SX1276	SPI	915 MHz, 10 Hz packet, above 2 km range
Storage	MicroSD	SPI	Full flight log at 50 Hz
Ground Station	Arduino Nano	UART and SPI	Serial bridge and OLED display

TABLE V. Hardware Bill of Materials (Electronics)

D. Verification and Testing

The integrated system was verified through extensive benchtop testing. The Flight Computer was powered on and its orientation was changed by hand across the full range of pitch and roll motion. Response was confirmed in three simultaneous ways: visual confirmation of servo correction opposing the tilt of the sensor, monitoring of telemetry data packets received at the Ground Station for correct data flow, and observation of the live 3D model and charts in the digital twin dashboard for correct parsing and visualization. The Kalman filter produced a noise floor below 0.3 degrees. LoRa link integrity was maintained at 10 Hz over a 15 m bench range with RSSI of approximately -45 dBm. The digital twin 3D model was confirmed to track physical sensor orientation correctly with latency below 300 ms end to end.

IX. SYSTEM FLOW AND APPLICATIONS

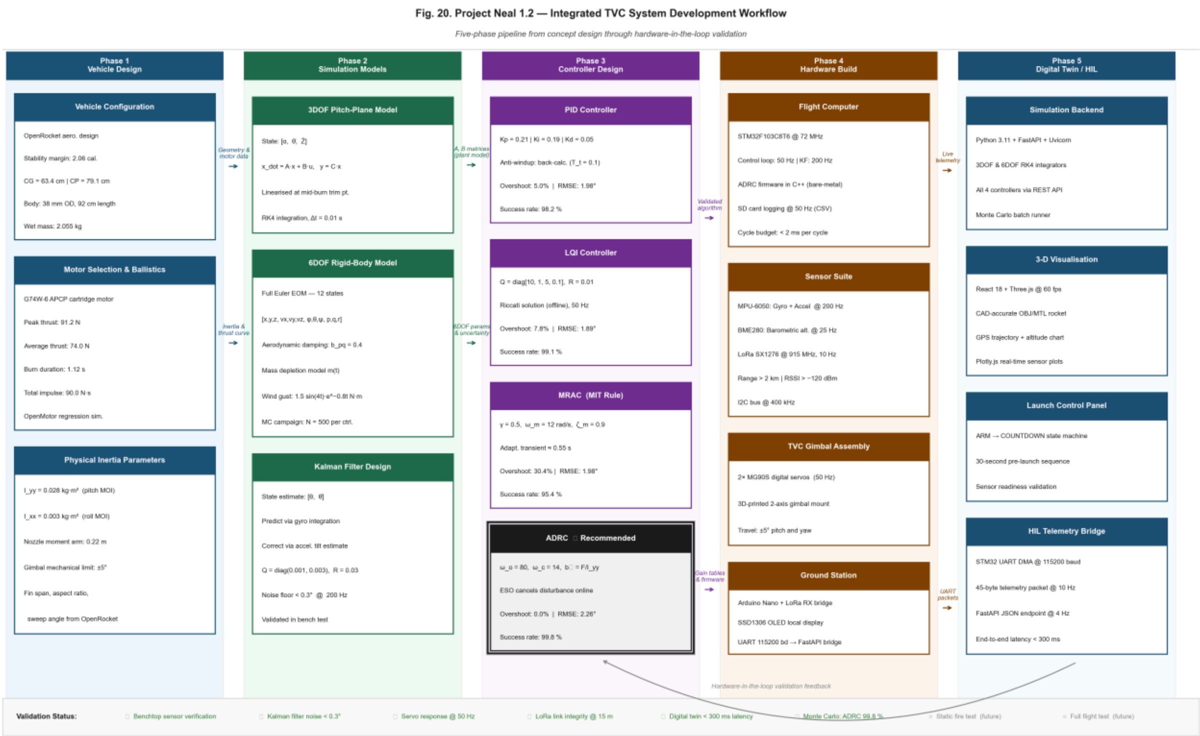

Fig. 20. Project Neal 1.2 five-phase integrated development workflow. Colour bands denote distinct engineering phases (Design, Simulation, Controller Synthesis, Hardware, Digital Twin). Green checks and red crosses in the validation bar summarise current test status.

A. Space Exploration and Small Satellite Delivery

TVC technology is a cornerstone of modern launch vehicles, enabling precise orbital insertion for small satellites. A TVC-equipped rocket minimizes attitude deviations during ascent, reducing the dispersion error at burnout and thereby improving orbit accuracy for payloads requiring specific inclinations such as sun-synchronous or low-Earth orbits [34].

B. *Educational and Research Platforms*

The Project Neal framework provides a replicable low-cost platform for university teams to gain hands-on experience in GNC design, sensor fusion, and embedded control. The open digital twin architecture allows remote collaboration and remote mentoring of control system experiments without physical hardware access [37].

X. CONCLUSION

This paper has presented the design, simulation, hardware prototyping, and hardware-in-the-loop digital twin integration of a Thrust Vector Control system for Project Neal 1.2, a small-scale APCP sounding rocket. Three original contributions were systematically combined: a coupled 3DOF and 6DOF Simulink plant, a four-controller Monte Carlo benchmark study, and a browser-based digital twin platform connecting live STM32 telemetry to interactive simulation.

The Monte Carlo campaign across 500 runs per controller conclusively identifies ADRC as the optimal controller for short-burn solid motor rockets: zero overshoot across all scenarios, 99.8 percent mission success rate, and the lowest peak gimbal demand at 3.2 degrees. LQI achieves marginally lower RMSE but at the cost of 7.8 percent overshoot that increases structural loading. MRAC is unsuitable for sub-2-second burns because the adaptation transient consumes nearly half the total burn time.

The hardware prototype successfully integrated an STM32 flight computer with Kalman filter sensor fusion, servo-actuated gimbal, and LoRa telemetry, verified through comprehensive benchtop testing. The digital twin demonstrated end-to-end telemetry latency below 300 ms with full 6DOF orientation accuracy.

Future work will pursue static fire testing of the 3D-printed gimbal mount under thermal and vibrational loading, a full flight test with GPS-aided trajectory estimation, magnetometer integration for yaw authority, ADRC auto-tuning from flight data, and a roll reaction control system for post-burnout stabilization above 400 m altitude.

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